

SESSION 1 — Introduction to Statistics

(Descriptive vs Inferential Statistics)

What is Statistics?

Definition

Statistics is the science of:

Collecting, organizing, analyzing, and interpreting data to make decisions.

In simple words:

Statistics helps us **convert raw data into useful information**.

Real-World Examples

1. Netflix analyzes watch-time data to recommend movies
 2. Amazon studies customer purchases to suggest products
 3. Hospitals analyze patient data to predict disease
 4. Banks analyze transactions to detect fraud
 5. Governments analyze census data for planning
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For a Data Analyst

Statistics helps you:

Understand data

Summarize large datasets

Find patterns

Make predictions

Take business decisions

Types of Statistics

There are **TWO** main types:

A) Descriptive Statistics

Definition

Descriptive statistics **describe and summarize** data.

It answers:

What happened?

Examples of Descriptive Stats

- Mean (average)
 - Median
 - Mode
 - Standard deviation
 - Charts (bar chart, pie chart, histogram)
-

Real Example

A company records daily sales:

100, 120, 130, 150, 200

Descriptive stats:

- Mean = 140
- Max = 200
- Min = 100

It only **describes past data**.

Data Analyst Use

Dashboards

KPI reports

Monthly sales reports

Customer summary

Excel / Power BI charts

B) Inferential Statistics

Definition

Inferential statistics **draw conclusions about a population using a sample**.

It answers:

What will happen?

Is this change real or by chance?

Examples

- Hypothesis testing
 - Confidence intervals
 - t-test
 - Regression
 - ANOVA
-

Real Example

You test a **new price** on 100 customers (sample)
You conclude it will work for **all customers (population)**.

Data Analyst Use

A/B testing
Marketing impact analysis
Churn prediction
Revenue forecasting
Product experiments

Population vs Sample

Population

Entire group of interest
Example:
All customers of Amazon = population

Sample

Small subset taken from population
Example:
1000 customers selected from Amazon users

Why sample?

- Cheaper
 - Faster
 - Sometimes population is too large
 - Sometimes impossible to measure whole population
-

Example

Population = all Indian voters
Sample = 5000 voters surveyed

Parameter vs Statistic

Concept	Meaning	Symbol
Parameter	Measure from population	μ (mu), σ
Statistic	Measure from sample	$\bar{x}$ (mean), s

Example

Population average salary = μ = ₹50,000

Sample average salary = $\bar{x}$ = ₹48,000

Business Use of Statistics

Companies use statistics to:

Forecast sales
 Improve quality
 Reduce risk
 Understand customers
 Detect fraud
 Optimize pricing

Excel Demo (Concept)

Dataset: Customer purchases

Columns:

CustomerID | Spend

Steps:

1. Treat all data = population
2. Randomly pick 20 rows = sample
3. Compare:
 - Mean of population
 - Mean of sample
4. Observe difference

Math Formulas

Sample Mean:

$$\bar{x} = \frac{\sum x}{n} \quad \bar{x} = \frac{\sum x}{n}$$

Population Mean:

$$\mu = \frac{\sum X}{N} \quad \mu = \frac{\sum X}{N}$$

Where:

x = sample values

X = population values

n = sample size

N = population size

Examples

Example 1

Sales of 5 days: 100, 120, 130, 150, 200

Mean = $(700)/5 = 140$

Example 2

Company surveys 100 customers out of 10,000

100 = sample

10,000 = population

Example 3

Average height of class = statistic

Average height of all students in college = parameter

Example 4

You analyze last year's revenue (descriptive)

You predict next year's revenue (inferential)

Example 5

Churn rate from sample = 5%

Estimated churn of population = 5%

Mini Project – Session 1

Project: Customer Sales Summary

Dataset (create in Excel):

CustomerID | Sales

Tasks:

1. Treat full data as population
2. Take random 10 customers as sample

3. Compute:
 - Population mean
 - Sample mean
 4. Compare both
 5. Write 3 insights:
 - Is sample close to population?
 - Is sampling reliable?
 - Business decision
-

10 Questions

Theory (5)

1. Define statistics : It is the science of collecting, organizing, analyzing and interpreting raw data into useful insights.
 2. Difference between descriptive and inferential statistics : Descriptive stats tells us what happened from the past data, and inferential stats predicts what will happen from a small selected data.
 3. What is population? : In inferential stats the overall data taken is known as population.
 4. What is sample? : It is a small part of the population used to do inferential stats.
 5. Difference between parameter and statistic : The answer we get from population is known as parameter and the answer from sample is known as statistic.
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Practical (5)

1. Identify population and sample: Survey of 200 voters out of 50,000
Population : 50000, Sample : 200
2. Sales data of last year → descriptive or inferential? : descriptive because data from past
3. Predicting next month sales → which type? : Inferential because prediction
4. Mean of sample = 500, population mean = 520 → which is parameter? : 520
5. Give one business example of inferential statistics : beta testing in games.

SESSION 2 — Types of Data & Measurement Scales

What Do We Mean by “Types of Data”?

Definition

Data types tell us:

- What kind of values a column contains
- What calculations are allowed
- Which charts, tests, and models we can use

Choosing the **wrong data type** = **wrong analysis**

Major Classification of Data

There are **two broad types of data**:

A) Numerical (Quantitative Data)

B) Categorical (Qualitative Data)

A) Numerical (Quantitative Data)

Definition

Numerical data consists of **numbers** on which mathematical operations like addition, subtraction, average, etc., make sense.

It answers:

- How many?
 - How much?
-

Types of Numerical Data

1. Discrete Data

Definition

Discrete data consists of **countable values**.
It is obtained by **counting**.

Key Property

No intermediate values are possible.

Examples

- Number of customers = 120
- Number of orders = 45
- Number of employees = 300
- Number of defective products = 7

You cannot have **2.5 customers**

Real-World Use Case

- Bank: number of transactions per day
 - Call center: number of calls received
 - Factory: number of items produced
-

For a Data Analyst

- Used in **bar charts**
 - Used in **frequency tables**
 - Used in **Poisson & Binomial distributions**
-

2. Continuous Data

Definition

Continuous data can take **any value within a range**, including decimals.

It is obtained by **measurement**.

Examples

- Salary = 45,678.75
 - Height = 170.4 cm
 - Time = 2.35 hours
 - Temperature = 36.6°C
-

Real-World Use Case

- Daily revenue
- Customer spending

- Sensor readings
 - Delivery time
-

For a Data Analyst

- Used in **histograms**
 - Used in **Normal distribution**
 - Used in **regression analysis**
-

B) Categorical (Qualitative Data)

Definition

Categorical data represents **labels or groups**, not numeric magnitude.

It answers:

- Which type?
-

Types of Categorical Data

3. Nominal Data

Definition

Nominal data has **categories with no natural order**.

Examples

- Gender (Male, Female)
 - Blood group (A, B, AB, O)
 - Country (India, USA, UK)
 - Product category (Electronics, Clothing)
-

Key Rule

No ranking

No average

For a Data Analyst

- Used in **pie charts**
 - Used in **bar charts**
 - Used in **Chi-square test**
-

4.Ordinal Data

Definition

Ordinal data has **categories with order**, but the difference between them is **not measurable**.

Examples

- Customer satisfaction (Low, Medium, High)
 - Education level (School, College, PG)
 - Hotel ratings (1-star to 5-star)
-

Key Rule

Order matters

Mean is not meaningful

For a Data Analyst

- Used in **survey analysis**
 - Used in **ranking problems**
 - Used in **non-parametric tests**
-

3. Time-Series Data

Definition

Time-series data is collected **over time** at regular intervals.

Examples

- Daily sales
- Monthly profit
- Yearly population

- Hourly website traffic
-

For a Data Analyst

- Trend analysis
 - Forecasting
 - Seasonality detection
-

4. Measurement Scales

Measurement scales define **what operations are allowed** on data.

There are **four scales**:

1. Nominal Scale

- Labels only
- No order

Examples: Gender, City, Product type

Frequency count

Mean, median

2. Ordinal Scale

- Ordered categories

Examples: Ratings, Satisfaction level

Median

Mean

3. Interval Scale

- Equal intervals
- No true zero

Examples: Temperature (°C), Calendar years

Addition, subtraction

Ratios

4. Ratio Scale

- Equal intervals

- True zero exists

Examples: Salary, Age, Sales
All arithmetic operations

Business Dataset Example

Telecom Dataset Columns

Column	Data Type	Scale
CustomerID	Categorical	Nominal
Gender	Categorical	Nominal
Tenure	Numerical	Ratio
MonthlyCharges	Numerical	Ratio
Contract	Categorical	Nominal
Churn	Categorical	Nominal

Correct classification helps choose:

- Charts
- Statistical tests
- ML models

Five Worked Examples

Example 1

Number of students → Discrete numerical

Example 2

Height of students → Continuous numerical

Example 3

Education level → Ordinal

Example 4

Temperature in Celsius → Interval

Example 5

Monthly income → Ratio

Mini Project — Session 2

Project: Customer Data Classification

Dataset (Excel):

- CustomerID
- Age
- Gender
- City
- SatisfactionLevel
- MonthlySpend

Tasks

1. Identify numerical vs categorical
 2. Identify discrete vs continuous
 3. Identify nominal vs ordinal
 4. Identify measurement scale
 5. Suggest best chart for each column
 6. Write 5 business insights
-

10 Practice Questions

Theory (5)

1. Define numerical data : They are the countable values on which arithmetic operations can be performed
 2. Difference between discrete and continuous : Discrete data contains whole number values and continuous data contains point values
 3. What is nominal data? : They are the categorized data with no order
 4. Difference between ordinal and interval : Ordinal data has categorized data with order but cannot be measured and interval data are non zero data with equal intervals.
 5. What is ratio scale? : Ratio scale is same as interval but here true zero value exist.
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Practical (5)

1. Classify: number of website visits : Discrete
2. Classify: customer rating (Poor–Excellent) : Ratio scale, ordinal
3. Classify: temperature in Celsius : Interval scale

4. Classify: blood group : Nominal data
5. Classify: annual income : time series data

SESSION 3 — Measures of Central Tendency

(Mean, Median, Mode)

What Are Measures of Central Tendency?

Definition

Measures of central tendency describe **the center or typical value** of a dataset.

They answer:

What is a “normal” or “representative” value?

The three most important measures are:

- Mean
- Median
- Mode

Mean (Arithmetic Average)

Definition

The **mean** is the sum of all values divided by the total number of observations.

Formula

$$\text{Mean} = \bar{x} = \frac{\sum x}{n} \quad \text{Mean} = \bar{x} = \frac{\sum x}{n}$$

Where:

- x = individual values
- n = number of observations

Example 1

Daily sales (₹):

100, 120, 130, 150, 200

$$\text{Mean} = \frac{100+120+130+150+200}{5} = 140$$

Example 2

Monthly revenue (₹ in thousands):

50, 52, 55, 60

Mean = 54.25

Real-World Use Case

- Average salary in a company
 - Average daily sales
 - Average customer spend
 - Average delivery time
-

Sensitivity to Outliers

Mean is **highly affected by extreme values**.

Example:

Salaries:

25k, 30k, 35k, 40k, **5,00,000**

Mean becomes misleading.

For a Data Analyst

- Used in KPI reports
 - Used in dashboards
 - Used in regression and ML models
 - Not suitable when outliers exist
-

Median (Middle Value)

Definition

The **median** is the **middle value** when data is arranged in ascending order.

Rules

- If n is odd $\rightarrow$ middle value
 - If n is even $\rightarrow$ average of two middle values
-

Example 3 (Odd)

Sales:

100, 120, 130, 150, 200

Median = 130

Example 4 (Even)

Sales:

100, 120, 130, 150

Median = $(120 + 130)/2 = 125$

Real-World Use Case

- Median income
 - Median house price
 - Median delivery time
 - Median customer spend
-

Advantage

Median is **robust to outliers**.

Example:

25k, 30k, 35k, 40k, **5,00,000**

Median = 35k (realistic)

For a Data Analyst

- Used in income analysis
 - Used in real-estate analytics
 - Used in skewed datasets
 - Preferred when data has outliers
-

Mode (Most Frequent Value)

Definition

The **mode** is the value that appears **most frequently**.

Example 5

Product sales:
Mobile, Laptop, Mobile, Tablet, Mobile
Mode = Mobile

Numerical Example

Data:
2, 3, 3, 4, 5
Mode = 3

Real-World Use Case

- Most sold product
 - Most common customer age
 - Most frequent complaint type
-

For a Data Analyst

- Useful for categorical data
 - Helps identify popular items
 - Used in market basket analysis
-

Comparison: Mean vs Median vs Mode

Measure Best Used When		Outlier Effect
Mean	Data is symmetric	High
Median	Data is skewed	Low
Mode	Categories or popularity	None

When to Use What?

Use Mean when:

- Data is evenly distributed
- No extreme outliers
- Example: exam scores

Use Median when:

- Data is skewed
- Outliers exist
- Example: income

Use Mode when:

- Data is categorical
- You want popularity
- Example: product preference

Business Demo Idea (Excel)

Monthly Sales Example

Sales (₹):

10k, 12k, 13k, 15k, **1,00,000**

- Mean = High (misleading)
- Median = Realistic
- Mode = May not exist

Insight: **Median gives better business picture**

Mini Project — Session 3

Project: Sales Performance Analysis

Dataset:

- Month
- Sales Amount

Tasks

1. Calculate mean sales
2. Calculate median sales
3. Identify mode (if any)
4. Add one outlier value
5. Compare before & after
6. Write 5 business insights

10 Practice Questions

Theory (5)

1. Define mean : It is the average of all data, calculated by adding all values and then dividing it by number of total values
 2. Define median : It is the center value of all values, if odd the centermost and if even then the mean of 2 centermost values.
 3. Define mode : It is calculated by seeing the maximum number of times occurring values.
 4. Why is mean affected by outliers? : Because outliers increase the broadness of data which affects mean.
 5. When is median preferred over mean? : Median is accurate then mean because it is the center value of asc or desc values and does not depend on outliers or any big values.
-

Practical (5)

1. Find mean of: 10, 20, 30, 40 : 25
2. Find median of: 5, 10, 15 : 10
3. Find median of: 5, 10, 15, 20 : 12.5
4. Find mode of: 2, 3, 3, 4 : 3
5. Identify best measure for income data : Median is the best because it ranges the data values and find its middle value.

SESSION 4 — Measures of Dispersion

(Range, Variance, Standard Deviation, Coefficient of Variation)

What Are Measures of Dispersion?

Definition

Measures of dispersion describe **how spread out or scattered** the data values are around the center (mean or median).

They answer:

How consistent is the data?

How much do values vary from each other?

Two datasets can have the **same mean** but **very different variability**.

Range

Definition

The **range** is the difference between the **maximum and minimum** value in a dataset.

Formula

Range = Max - Min

Example 1

Customer spend (₹):

200, 300, 400, 800

Range = 800 - 200 = 600

Real-World Use Case

- Price range of products
 - Temperature variation in a city
 - Salary range in a company
-

Limitation

Range depends only on **two values** and ignores the rest of the data.

For a Data Analyst

- Quick variability check
 - Used in early data exploration
 - Not reliable for final analysis
-

Variance

Definition

Variance measures the **average squared deviation** of each data point from the mean.

It tells us **how far data points are from the average**.

Formula

Population Variance

$$\sigma^2 = \frac{1}{N} \sum (X - \mu)^2$$

Sample Variance

$$s^2 = \frac{1}{n-1} \sum (x - \bar{x})^2$$

Example 2

Customer spend (₹):

10, 12, 14

Step 1: Mean = 12

Step 2: Deviations:

$$(10 - 12)^2 = 4$$

$$(12 - 12)^2 = 0$$

$$(14 - 12)^2 = 4$$

Step 3: Variance = $(4 + 0 + 4) / 3 = 2.67$

Limitation

Variance is expressed in **squared units**, which is hard to interpret.

For a Data Analyst

- Used internally in statistics
- Base for standard deviation

- Used in regression and ML algorithms
-

Standard Deviation

Definition

The **standard deviation (SD)** is the **square root of variance**.

It measures how much data typically deviates from the mean.

Formula

$SD = \sqrt{\text{Variance}}$

Example 3

Variance = 2.67

$SD = \sqrt{2.67} \approx 1.63$

Real-World Use Case

- Volatility in stock prices
 - Risk in investment returns
 - Consistency of manufacturing quality
 - Customer spending behavior
-

Interpretation

- Low SD → data is consistent
 - High SD → data is spread out
-

For a Data Analyst

- Used in performance analysis
 - Used in anomaly detection
 - Used in forecasting and risk analysis
-

Coefficient of Variation (CV)

Definition

The **coefficient of variation** measures **relative variability**.

It tells us how large the SD is **relative to the mean**.

Formula

$$CV = (SD / \text{Mean}) \times 100$$

Example 4

Mean spend = 500

SD = 50

$$CV = (50 / 500) \times 100 = 10\%$$

Real-World Use Case

- Compare variability between datasets
 - Compare risk across investments
 - Compare sales stability across regions
-

For a Data Analyst

- Useful for comparison
 - Helps rank consistency
 - Used in financial analytics
-

Five Worked Examples

Example 1

Sales data: 100, 110, 120

Low SD → stable sales

Example 2

Sales data: 50, 200, 500

High SD → unstable sales

Example 3

Two stores with same average sales

Store A: low SD → reliable

Store B: high SD → risky

Example 4

Product price variation analysis using CV

Example 5

Customer spend dispersion analysis

Business Demo Idea (Excel)

Dataset: Customer spending

Steps:

1. Calculate mean
 2. Calculate variance
 3. Calculate SD
 4. Compare two customer segments
 5. Identify which segment is more consistent
-

Mini Project — Session 4

Project: Customer Spending Consistency

Dataset:

- CustomerID
- MonthlySpend

Tasks:

1. Calculate range
 2. Calculate variance
 3. Calculate SD
 4. Calculate CV
 5. Identify high-variability customers
 6. Write 5 business insights
-

10 Practice Questions

Theory (5)

1. Define dispersion : It tells us how scattered the values are from the mean .
 2. What is range? : It is the difference between the max and min value of the data.
 3. Difference between variance and SD : Variance is the square of SD.
 4. Why is variance squared? : To remove all the negative values which might act as outliers and calculate the dispersion accurately.
 5. What does CV measure? : It measures how far is SD from mean. Lower the SD lower is the CV and higher is the accuracy of the data.
-

Practical (5)

1. Find range of: 10, 20, 50 : 40
2. Calculate variance of: 2, 4, 6 : 2.67
3. Calculate SD if variance = 9 : 3
4. Compare two datasets using CV
5. Identify which dataset is more stable

SESSION 5 — Percentiles, Quartiles & Boxplots

Outlier detection | Salary analysis | Customer segmentation | Business decision making

Percentiles

A **percentile** tells us the value below which a given percentage of observations fall.

-> It divides data into **100 equal parts**.

Conceptual Formula

Position of percentile:

$$P_k = k/100 \times (n+1)$$

Where:

- k = percentile number
 - n = number of observations
-

Example 1

Test scores:

50, 60, 70, 80, 90

50th percentile = Median = 70

90th percentile $\approx$ 90

Interpretation:

90% of students scored below 90.

Real-World Use Cases

- 90th percentile income
 - 95th percentile response time
 - Top 10% customers
 - Top-performing employees
-

For Data Analysts

- Used in customer ranking
- Used in salary benchmarking
- Used in performance analytics

- Used in ML feature scaling
-

Quartiles

Quartiles divide data into **4 equal parts**.

- Q1 → 25th percentile
 - Q2 → 50th percentile (Median)
 - Q3 → 75th percentile
-

Example 2

Data:

10, 20, 30, 40, 50, 60, 70

Q2 (Median) = 40

Lower half: 10, 20, 30

Q1 = 20

Upper half: 50, 60, 70

Q3 = 60

Interpretation

- Q1 → bottom 25%
 - Q2 → middle
 - Q3 → top 25%
-

Real-World Example

Customer spending:

Q1 = 500

Q3 = 2000

Top 25% customers spend more than 2000.

Interquartile Range (IQR)

IQR measures the spread of the middle 50% of data.

$IQR = Q3 - Q1$

Example 3

If:

$$Q1 = 20$$

$$Q3 = 60$$

$$IQR = 60 - 20 = 40$$

Why IQR Is Important

IQR helps detect **outliers**.

Outlier Rule:

Lower bound: $Q1 - 1.5 \times IQR$

Upper bound: $Q3 + 1.5 \times IQR$

Any value outside this range = Outlier.

Example 4

$$Q1 = 20$$

$$Q3 = 60$$

$$IQR = 40$$

$$\text{Lower bound} = 20 - 60 = -40$$

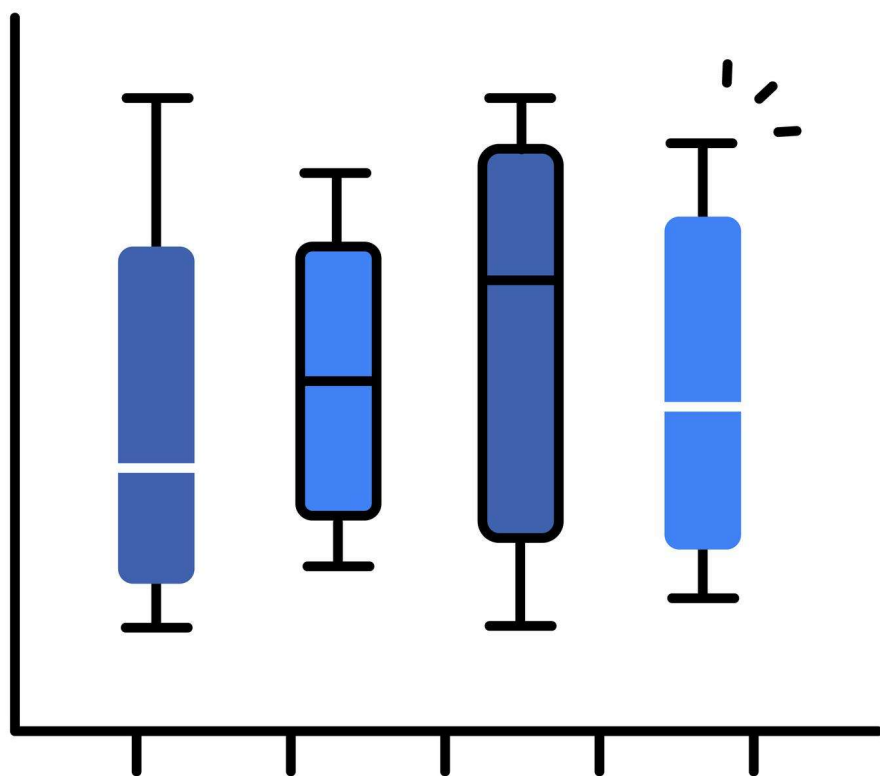
$$\text{Upper bound} = 60 + 60 = 120$$

Values outside $(-40, 120)$ are outliers.

Boxplot (Box-and-Whisker Plot)

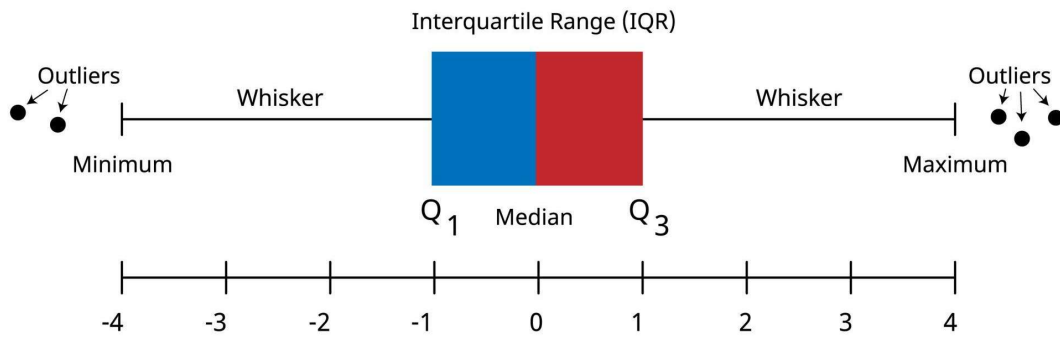
A **boxplot** visually shows:

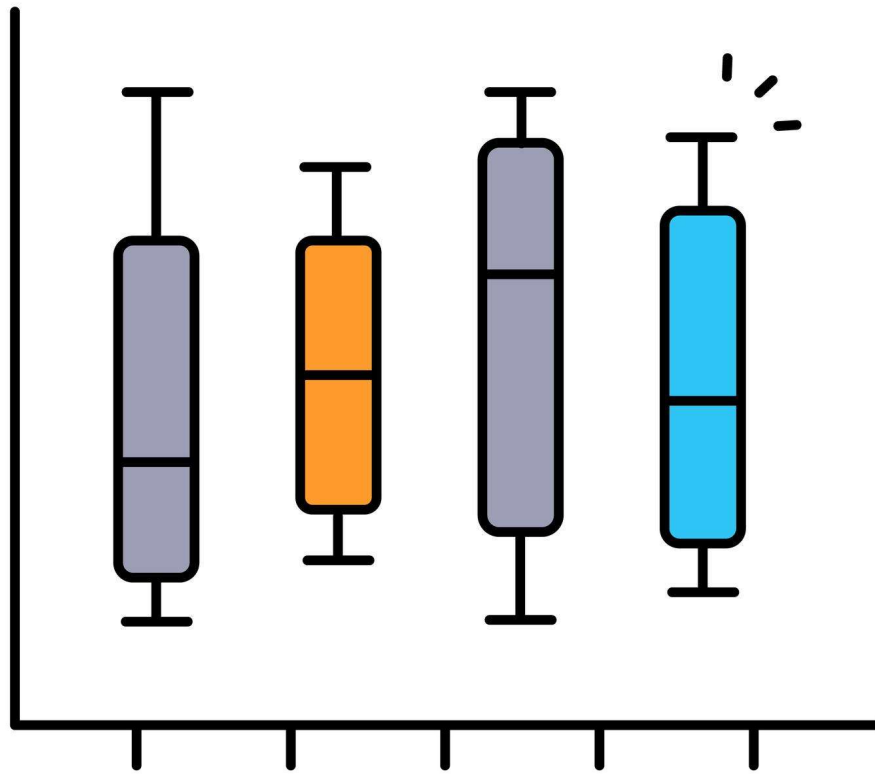
- Minimum
 - Q1
 - Median
 - Q3
 - Maximum
 - Outliers
-



Box Plot

Box and Whisker Plots





Box Plot

4

Parts of Boxplot

- Left whisker → Minimum
 - Left box edge → Q1
 - Line inside box → Median
 - Right box edge → Q3
 - Right whisker → Maximum
 - Dots outside → Outliers
-

Business Example

Company compares customer spending:

Segment A → tight box → consistent

Segment B → wide box → high variability

Five Worked Examples

Example 1

Find Q1, Q2, Q3 for:

5, 10, 15, 20, 25

Example 2

Identify top 10% customers using percentile

Example 3

Detect outliers in salary dataset

Example 4

Compare IQR of two stores

Example 5

Create boxplot of daily revenue

Business Demo (Python Idea)

Steps:

1. Load customer spend data
 2. Calculate percentiles using numpy
 3. Compute Q1, Q3, IQR
 4. Detect outliers
 5. Plot boxplot
-

Mini Project — Session 5

Project: Customer Spending Segmentation

Dataset:

CustomerID | MonthlySpend

Tasks:

1. Calculate Q1, Q2, Q3
2. Compute IQR
3. Detect outliers
4. Identify top 25% customers
5. Create boxplot
6. Write 5 business insights

10 Practice Questions

Theory (5)

1. What is percentile? : It ranks the data amongst all the other data.
2. Define quartiles : It is the process in which the total data in asc or desc order is divided into 4 equal parts.
3. What is IQR? : After dividing the data into quartiles, the middle two data is used to find the IQR value by subtracting them.
4. Why is IQR useful? : It helps to remove outliers and measure data accurately.
5. What does a boxplot show? : It shows the all values of the data such as the quartiles, IQR, max, min, outliers.



Practical (5)

1. Find Q1, Q2, Q3 of: 2, 4, 6, 8, 10 : Q1: 3, Q3: 9, Q2: 6
2. Calculate IQR : $Q3 - Q1 = 9 - 3 = 6$
3. Identify outlier using $1.5 \times \text{IQR}$ rule : Lower bound = $q1 - 1.5 * \text{IQR} = -6$, Upper bound = $q3 + 1.5 * \text{IQR} = 18$, any values outside -6 to 18 is outlier
4. Find 90th percentile of dataset : $P(90) = 90(5+1) / 100 = 5.4 = 5^{\text{th}}$ position , so in dataset it is 10
5. Interpret a boxplot : It shows the all values of the data such as the quartiles, IQR, max, min, outliers.

SESSION 6 — Probability Basics

Probability is the backbone of:

- Machine Learning
- Risk analysis
- Forecasting
- Decision science
- A/B testing

What is Probability?

Probability measures the likelihood that an event will occur.

It ranges between:

$$0 \leq P(E) \leq 1$$

- 0 → Impossible event
- 1 → Certain event

Example

- Probability of sun rising tomorrow ≈ 1
- Probability of getting 7 on a dice = 0

Real-World Business Use

- Probability customer will churn
- Probability loan default
- Probability product defect
- Probability fraud

Key Terms in Probability

Experiment

An action with uncertain outcome.

Examples:

- Tossing a coin
- Rolling a dice
- Customer visiting website

Outcome

Single possible result of an experiment.

Example:

Dice roll $\rightarrow 3$

Sample Space (S)

All possible outcomes.

For dice:

$S=\{1,2,3,4,5,6\}$

Event (E)

Subset of sample space.

Example:

Getting even number:

$E=\{2,4,6\}$

Classical Probability

If outcomes are equally likely:

$P(E)=\text{Number of favorable outcomes} / \text{Total outcomes}$

Example 1 — Dice

Probability of getting 4:

$P(4)=1/6$

Example 2 — Coin

Sample space:

$\{H,T\}$ $P(H)=1/2$

Example 3 — Business Case

100 customers

30 buy Product A

$P(\text{Buy A})=30/100=0.3$

Addition Rule of Probability

Used when finding probability of:

Event A OR Event B

Formula

If A and B are mutually exclusive:

$$P(A \cup B) = P(A) + P(B)$$

If not mutually exclusive:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example 4

Probability customer buys:

- Product A = 0.4
- Product B = 0.3
- Both = 0.1

$$P(A \cup B) = 0.4 + 0.3 - 0.1 = 0.6$$

Conditional Probability (Basic Idea)

Probability of A given B happened.

$$P(A|B)$$

Formula

$$P(A|B) = P(A \cap B) / P(B)$$

Example 5

100 customers

40 bought accessories

20 bought phone AND accessories

$$P(\text{Phone} | \text{Accessories}) = 20/40 = 0.5$$

Interpretation:

If customer bought accessories, there is 50% chance they buy phone.

Real Business Example — E-Commerce

Suppose:

- 1000 customers

- 300 buy Product A
- 200 buy Product B
- 100 buy both

Find probability customer buys A OR B:

$$0.3+0.2-0.1=0.4$$

So 40% customers buy at least one.

Five Practical Examples

Example 1

Probability of red card from deck

Example 2

Probability customer churn if 50 out of 200 churn

Example 3

Probability of at least one success

Example 4

Probability of even number on dice

Example 5

Probability user clicks ad

Business Demo Idea (Excel)

Dataset:

CustomerID | BoughtProductA | BoughtProductB

Tasks:

1. Count customers buying A
2. Count customers buying B
3. Count customers buying both
4. Compute:
 - $P(A)$
 - $P(B)$

- $P(A \text{ OR } B)$
- $P(A|B)$

Mini Project — Session 6

Project: Product Purchase Probability

Dataset:

CustomerID | PurchasedPhone | PurchasedAccessories

Tasks:

1. Calculate $P(\text{Phone})$
2. Calculate $P(\text{Accessories})$
3. Calculate $P(\text{Phone OR Accessories})$
4. Calculate $P(\text{Phone} | \text{Accessories})$
5. Write 5 business insights

10 Practice Questions

Theory (5)

1. Define probability : It is likelihood of the event to occur.
2. What is sample space? : It is the all possible outcomes.
3. What is an event? : It is a subset of sample space which fulfils specific condition.
4. State addition rule : It is the probability of event A or event B.
5. Define conditional probability : It is the probability of an event A given that event B has occurred.

Practical (5)

1. Find probability of rolling 3 on dice : $1/6$
2. Find probability of even number on dice : $3/6 = 1/2$
3. 60 out of 200 customers churn. Find probability : 0.3
4. Calculate $P(A \text{ OR } B)$ if given $P(A)$, $P(B)$, $P(A \cap B)$: $P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$
5. Compute conditional probability example : $P(A|B) = P(A \cap B) / P(B)$

SESSION 7 — Conditional Probability & Bayes' Theorem

This session is the backbone of:

- Churn prediction
- Fraud detection
- Spam filtering
- Medical diagnosis
- Machine learning classification

Conditional Probability (Deep Understanding)

Conditional probability measures the probability of an event **given that another event has already occurred**.

We write it as:

$$P(A|B)$$

Read as:

Probability of A **given** B.

Formula

$$P(A|B) = P(A \cap B) / P(B)$$

Where:

- $P(A \cap B)$ = probability of both A and B
- $P(B)$ = probability of B

Example 1 — Basic Dice Example

What is probability of getting even number given that number is greater than 3?

Sample space:

$\{1, 2, 3, 4, 5, 6\}$

$B = \text{greater than } 3 \rightarrow \{4, 5, 6\}$

$A = \text{even} \rightarrow \{2, 4, 6\}$

$A \cap B = \{4, 6\}$

$$P(A|B) = 2/3$$

Business Example — E-Commerce

Suppose:

- 1000 customers

- 400 bought accessories
- 200 bought phone
- 150 bought both

Find:

$$P(\text{Phone} \mid \text{Accessories}) = 150 / 400 = 0.375$$

Interpretation:

If customer bought accessories, there is 37.5% chance they buy phone.

Why This Matters for Data Analysts

This helps in:

- Cross-selling
- Recommendation systems
- Customer targeting
- Association rule mining

Bayes' Theorem

Now we go one level deeper.

Bayes' Theorem helps us **reverse conditional probability**.

Instead of:

$$P(A|B)$$

We calculate:

$$P(B|A)$$

Bayes' Formula

$$P(A|B) = P(B|A) \cdot P(A) / P(B)$$

Where:

- $P(A)$ = Prior probability
- $P(B|A)$ = Likelihood
- $P(B)$ = Evidence
- $P(A|B)$ = Posterior probability

Medical Diagnosis Example

Suppose:

- 1% population has disease
- Test accuracy:
 - 95% detects disease if present
 - 5% false positive

Find probability that person actually has disease given test is positive.

Let:

A = Has disease

B = Test positive

Given:

$$P(A) = 0.01$$

$$P(B|A) = 0.95$$

$$P(B|\neg A) = 0.05$$

First find $P(B)$:

$$P(B) = P(B|A)P(A) + P(B|\neg A)P(\neg A)$$

$$= (0.95)(0.01) + (0.05)(0.99)$$

$$= 0.0095 + 0.0495 = 0.059$$

Now apply Bayes:

$$P(A|B) = 0.95 \times 0.01 / 0.059 = 0.161$$

Final Answer:

Even if test is positive, probability of actual disease is only 16.1%.

Important Insight

Base rate (prior probability) matters a lot.

This is called **Base Rate Fallacy**.

Fraud Detection Example

Suppose:

- 2% transactions are fraud
- Model detects fraud correctly 90%
- False positive rate 5%

You can compute probability that flagged transaction is actually fraud using Bayes.

This is exactly how:

- Credit card fraud systems work

- Spam filters work
- ML classifiers update probabilities

Five Worked Examples

Example 1

Probability customer churn given contract type

Example 2

Probability loan default given low credit score

Example 3

Probability customer buys product given email opened

Example 4

Medical testing probability

Example 5

Spam email detection probability

Business Demo Idea (Excel or Python)

Dataset:

CustomerID

BoughtAccessories

BoughtPhone

Tasks:

1. Calculate $P(\text{Phone})$
2. Calculate $P(\text{Accessories})$
3. Calculate $P(\text{Phone} \mid \text{Accessories})$
4. Reverse it using Bayes

Mini Project — Session 7

Project: Customer Purchase Probability Analysis

Dataset:

CustomerID | BoughtPhone | BoughtAccessories | HighIncome

Tasks:

1. Calculate conditional probabilities
2. Apply Bayes theorem
3. Identify high-probability segments
4. Recommend targeting strategy
5. Write 5 business insights

10 Practice Questions

Theory (5)

1. Define conditional probability : It is the probability to be found given one event condition
2. State formula of $P(A|B)$: $P(A|B) = P(A \cap B) / P(B)$
3. What is Bayes' theorem? : It is the reverse of conditional probability
4. What is prior probability? : It is the given probability
5. What is posterior probability? : It is the probability to be found

Practical (5)

1. Compute $P(A|B)$ if $P(A \cap B)=0.2$ and $P(B)=0.5$: ans = 0.4
2. Apply Bayes formula : $P(A|B)=P(B|A) \cdot P(A)/P(B)$
3. Solve simple medical test problem
4. Find fraud probability example
5. Interpret business conditional probability

SESSION 10 — Sampling Techniques

Sampling is extremely important because in real-world analytics we **rarely analyze the entire population**. Instead, we study a **sample** and draw conclusions about the population.

Why Sampling Is Required

Sampling is the process of selecting a **subset of individuals or observations from a population** to estimate characteristics of the whole population.

Real-World Problem

Imagine a company like **Amazon** with **200 million customers**.

It is impossible to analyze every customer for every study.

Instead we:

- Select a **representative sample**
- Analyze it
- Generalize findings to the population

Reasons for Sampling

1. Saves time
2. Reduces cost
3. Easier to analyze
4. Allows quick business decisions
5. Practical for very large datasets

Example

Population = 1,000,000 customers

Sample = 10,000 customers

We estimate:

- Average purchase value
- Churn probability
- Product preferences

Types of Sampling

The main sampling techniques used in analytics are:

1. Random Sampling
2. Stratified Sampling

3. Systematic Sampling
4. Cluster Sampling (advanced concept)

In this session we focus on the **most important three**.

Random Sampling

In **random sampling**, every member of the population has an **equal chance of being selected**.

Example

Population: 10,000 customers

Randomly select 500 customers.

Every customer has probability:

$$P = 500 / 10000$$

Business Example

A company sends a survey to randomly selected customers to measure satisfaction.

Advantages

- Reduces bias
- Easy to implement
- Statistically reliable

Limitations

May not represent subgroups properly.

Example:

- Only 10% female customers selected accidentally.

Stratified Sampling

In **stratified sampling**, the population is divided into **subgroups (strata)** and samples are taken from each group.

Example

Customer segments:

Segment	Customers
Students	4000

Segment	Customers
---------	-----------

Working professionals	4000
-----------------------	------

Senior citizens	2000
-----------------	------

If sample size = 1000

Then sample proportionally:

- Students = 400
- Professionals = 400
- Seniors = 200

Business Example

A telecom company studies **customer churn** by ensuring each segment is represented.

Advantages

- Better representation
- Higher accuracy
- Reduces sampling bias

Systematic Sampling

In **systematic sampling**, we select every **k-th observation** from a population list.

Formula

$$k = N / n$$

Where:

N = population size

n = sample size

Example

Population = 10,000 customers

Sample = 1,000

$$k = 10k / 1k = 10$$

Select every **10th customer**.

Business Example

A retailer like **Walmart** checks quality of products by inspecting every 50th item on the assembly line.

Advantages

- Simple to implement
- Efficient for large datasets

Limitation

If population has pattern → sampling bias.

Sample Size Importance

Choosing the correct sample size is critical.

Too small → inaccurate results

Too large → unnecessary cost

General Rule

Common sample sizes:

Population Sample Size

10,000 370

100,000 383

1,000,000 ~400

This is why surveys often sample **400–1000 respondents**.

Five Practical Examples

Example 1

Randomly select 500 website visitors for feedback.

Example 2

Stratify customers by region and sample proportionally.

Example 3

Select every 20th transaction for audit.

Example 4

Sample manufacturing products for quality testing.

Example 5

Sample social media posts for sentiment analysis.

Python Demo Idea

Steps:

1. Load customer dataset
2. Perform random sampling using pandas
3. Perform stratified sampling by customer segment
4. Perform systematic sampling
5. Compare results

Mini Project — Session 10

Project: Customer Behavior Sampling

Dataset columns:

CustomerID
Age
City
Segment
MonthlySpend

Tasks

1. Perform random sampling (10%)
2. Perform stratified sampling by segment
3. Perform systematic sampling
4. Compare mean spending of each sample
5. Identify which sampling method represents population best
6. Write 5 insights

10 Practice Questions

Theory (5)

1. Define sampling. : In real world we do not do analysis on data which sometimes may be very huge, so we create a sample of it and it is known as sampling.
2. Why is sampling needed? In real world we do not do analysis on data which sometimes may be very huge, so we create a sample of it (known as sampling) and do study on it.
3. What is random sampling? : in it every member of population has equal chance of being selected.
4. What is stratified sampling? : in it the population is divided into subgroups and samples are taken from each group.

5. What is systematic sampling? : in it we select every kth observation from the population, where $k = \text{population size}(N) / \text{sample size}(n)$.

Practical (5)

1. If population = 10,000 and sample = 1,000, find k for systematic sampling. : 10
2. Give a business example of stratified sampling. : a company checks quality of product by just taking every kth item.
3. Identify sampling type used in quality control.
4. Why is sample size important? : Sample size directly affects the **accuracy and reliability** of results. More the data more the accuracy.
5. How does random sampling reduce bias? : it ensures equal chance of being selected.

SESSION 12 — Hypothesis Testing Concepts

Hypothesis testing helps analysts **make decisions using data**.

It is widely used in:

- A/B testing
- Marketing experiments
- Product improvements
- Business decision making

Companies like **Google, Amazon, and Netflix** run thousands of hypothesis tests to determine whether product changes improve performance.

What Is Hypothesis Testing?

Hypothesis testing is a statistical method used to determine whether there is enough evidence in a sample dataset to conclude something about the population.

In simple words:

We test whether an assumption about data is **likely true or false**.

Key Components of Hypothesis Testing

There are two main hypotheses.

H_0 — Null Hypothesis

The **null hypothesis** represents the **default assumption**.

It usually states:

There is **no effect, no difference, or no relationship**.

Example:

New discount **does not increase sales**.

H₁ — Alternative Hypothesis

The **alternative hypothesis** represents the **claim we want to test**.

Example:

New discount **increases sales**.

Example Table

Hypothesis	Meaning
H ₀	Discount has no effect on sales
H ₁	Discount increases sales

Hypothesis Testing Process

Typical steps followed by data analysts:

1. Define null hypothesis
2. Define alternative hypothesis
3. Select significance level (α)
4. Compute test statistic
5. Calculate p-value
6. Decide whether to reject H₀

Significance Level (α)

The **significance level** is the probability of rejecting the null hypothesis when it is actually true.

Common values used in analytics:

Significance Level	Meaning
0.05	5% risk of false conclusion
0.01	1% risk

Most business experiments use:

$\alpha=0.05$

Type I and Type II Errors

Hypothesis testing can produce two types of errors.

Type I Error

Definition

Rejecting the null hypothesis **when it is actually true**.

This is also called **False Positive**.

Example:

You conclude the marketing campaign increased sales, but in reality it did not.

Probability of Type I error:

$$P(\text{Type I Error}) = \alpha$$

Type II Error

Definition

Failing to reject the null hypothesis **when it is false**.

This is also called **False Negative**.

Example:

The campaign actually increased sales, but your test failed to detect it.

Probability: β

Error Summary Table

Reality	Decision	Error Type
H_0 true	Reject H_0	Type I
H_0 false	Fail to reject H_0	Type II

p-value

The **p-value** is the probability of obtaining results **at least as extreme as the observed data**, assuming the null hypothesis is true.

Decision Rule

Condition	Decision
-----------	----------

$p\text{-value} \leq \alpha$	Reject H_0
------------------------------	--------------

$p\text{-value} > \alpha$	Fail to reject H_0
---------------------------	----------------------

Example

Suppose:

$$p\text{-value} = 0.03$$

$$\alpha = 0.05$$

Since:

$$0.03 < 0.05$$

We **reject the null hypothesis**.

Conclusion:

Discount likely increased sales.

Confidence Level

Confidence level is related to significance level.

Confidence Level = $1 - \alpha$

Common values:

α Confidence

0.05 95%

0.01 99%

Meaning:

We are **95% confident** in our conclusion.

Real Business Example

An e-commerce company runs a marketing experiment.

Old website conversion rate = 10%

After redesign:

Sample conversion rate = 12%

Hypothesis test:

H_0 : New design does not improve conversion

H_1 : New design improves conversion

If p-value < 0.05 $\rightarrow$ redesign is successful.

Five Worked Examples

Example 1

Testing whether a new product price increases revenue.

Example 2

Testing if average delivery time changed after logistics upgrade.

Example 3

Testing whether a marketing campaign increased customer signups.

Example 4

Testing whether a new training program improves employee productivity.

Example 5

Testing whether a new website layout increases click-through rate.

Python Demo Idea

Steps:

1. Load sales dataset
2. Calculate mean sales before campaign
3. Calculate mean sales after campaign
4. Perform hypothesis test
5. Compute p-value
6. Draw conclusion

Mini Project — Session 12**Project: Marketing Campaign Impact**

Dataset columns:

Date

SalesBeforeCampaign

SalesAfterCampaign

Tasks

1. Define H_0 and H_1 : H_0 is the null hypothesis which means that there is no change in default assumption and H_1 is the claim we want to test.
2. Compute average sales before and after
3. Perform hypothesis test
4. Calculate p-value
5. Decide whether campaign improved sales
6. Write 5 business insights

10 Practice Questions**Theory (5)**

1. What is hypothesis testing? : It is a statistical method which tells us whether the evidence from the sample is enough to tell about the population.

2. Define null hypothesis. : It means that there is no change in default assumption.
7. Define alternative hypothesis. : H_1 is the claim we want to test.
3. What is Type I error? : It means that we are rejecting the null hypothesis which is actually true.
4. What is p-value? : It is used in statistics in hypothesis testing to decide whether the result we found is significant(as extreme as the observed data) or just based on random data.

Practical (5)

1. If p-value = 0.02 and $\alpha = 0.05$, what is the decision? : We reject the null hypothesis
2. If $\alpha = 0.01$, what is the confidence level? : $1 - \alpha = 99\%$ confidence level
3. Give a business example of hypothesis testing. : Testing whether a new product price increases revenue.
4. What happens if p-value > α ? : we fail to reject the null hypothesis.
5. Define Type II error with example. : it is the failure of rejecting null hypothesis when it is false. The campaign increased sales but we failed to detect it.

SESSION 13 — t-Tests

(One-Sample, Two-Sample, Paired t-Test)

t-tests are used to **compare means** and determine whether differences are statistically significant.

They are widely used in:

- A/B testing
- Marketing experiments
- Product analysis
- Revenue comparison
- Before–after analysis

Companies like **Amazon** and **Netflix** frequently use t-tests to measure whether changes improve user behavior.

What Is a t-Test?

A **t-test** is a statistical test used to determine whether the **mean of a sample differs significantly from a population mean or from another sample mean**.

It is mainly used when:

- Sample size is small
- Population standard deviation is unknown

General t-Statistic Formula

$t = \text{Observed Difference} / \text{Standard Error}$

For most cases:

$$t = [\bar{x} - \mu] / [s / \sqrt{n}]$$

Where:

- $\bar{x}$ = sample mean
- μ = population mean
- s = sample standard deviation
- n = sample size

Types of t-Tests

Three major types are used in analytics:

One-Sample t-Test

Independent Two-Sample t-Test

Paired t-Test

One-Sample t-Test

Used when we compare **sample mean with a known population mean**.

Example

A company claims:

Average delivery time = 30 minutes.

A sample of 25 deliveries shows:

Mean delivery time = 33 minutes.

Hypothesis:

$H_0: \mu = 30$

$H_1: \mu \neq 30$

We perform a **one-sample t-test**.

Business Use Cases

- Test if average revenue changed
- Test if delivery time differs from target
- Test if average customer rating differs from benchmark

Independent Two-Sample t-Test

Used to compare the means of **two independent groups**.

Example

Compare average spending of:

Group A → Customers who saw advertisement

Group B → Customers who did not

Hypothesis:

H_0 : Mean_A = Mean_B

H_1 : Mean_A $\neq$ Mean_B

Business Use Cases

- A/B testing
- Compare two marketing strategies
- Compare two product versions

Paired t-Test

Used when the **same subjects are measured twice**.

Example:

Before vs After.

Example

Measure employee productivity:

Before training

After training

We test whether productivity increased.

Business Use Cases

- Before–after marketing campaign
- Before–after product redesign
- Before–after training program

Example Dataset

Before marketing campaign sales:

50, 55, 60, 62, 65

After campaign:

70, 75, 78, 80, 82

We perform a **paired t-test** to check if sales increased.

Visualization Idea

You can visualize:

- Boxplot before vs after
- Mean difference
- Distribution comparison

Five Worked Examples

Example 1

Test if average customer rating differs from 4.5.

Example 2

Compare average revenue between two marketing campaigns.

Example 3

Compare employee performance before and after training.

Example 4

Test if website redesign improves session duration.

Example 5

Test whether new delivery system reduces delivery time.

Mini Project — Session 13

Project: Marketing Campaign Effectiveness

Dataset columns:

CustomerID

RevenueBeforeCampaign

RevenueAfterCampaign

Tasks

1. Define H_0 and H_1
2. Calculate average revenue before and after
3. Perform paired t-test
4. Compute p-value
5. Interpret result

6. Visualize results
7. Write 5 business insights

10 Practice Questions

Theory (5)

1. What is a t-test? : It is a statistical test used to determine whether the mean of a sample differs significantly from a population mean or from another sample mean.
2. When is one-sample t-test used? : It is used when we compare sample mean with a known population mean.
3. What is independent t-test? : It is used to compare the means of two independent groups.
4. What is paired t-test? : It is used when the same subjects are measured twice.
5. Why are t-tests important in analytics? : It is important for A/B testing, Product analytics, before-after analysis.

Practical (5)

1. Write formula of t-statistic. : $t = [\bar{x} - \mu] / [s / \sqrt{n}]$
2. Identify correct t-test for A/B testing. : independent.
3. Explain paired t-test with example. Measure employee productivity before and after training to test whether productivity increased.
4. If p-value < 0.05, what is decision? : we reject the null hypothesis.
5. Give real business example using t-test. : Compare employee performance before and after training.

SESSION 14 — Chi-Square Test

(Goodness of Fit & Test of Independence)

The **Chi-Square Test (χ^2 Test)** is used to analyze **relationships between categorical variables**.

It is widely used in:

- Market research
- Customer preference analysis
- Survey analysis
- Demographic studies
- Product recommendation analysis

Companies like **Amazon** and **Netflix** analyze user behavior using categorical relationships such as **gender vs product preference** or **region vs subscription type**.

What Is the Chi-Square Test?

The **Chi-Square test** is a statistical test used to determine whether there is a **significant association between categorical variables**.

It compares:

Observed values vs Expected values.

Chi-Square Statistic Formula

$$\chi^2 = \sum [(O - E)^2 / E]$$

Where:

- O = Observed frequency
- E = Expected frequency

Interpretation

If the difference between observed and expected values is **large**, the χ^2 statistic becomes large → indicates relationship.

Types of Chi-Square Tests

There are two main types used in analytics.

Test	Purpose
Goodness of Fit	Check if data matches expected distribution
Test of Independence	Check relationship between two categorical variables

Chi-Square Goodness of Fit Test

Used to determine whether **observed data fits a theoretical distribution**.

Example

A company expects product sales to be evenly distributed among four categories.

Expected sales:

Product Expected

A	25
B	25
C	25
D	25

Observed sales:

Product Observed

A	30
B	20
C	28
D	22

We use **Chi-square goodness of fit** to check whether distribution differs significantly.

Chi-Square Test of Independence

Used to determine whether **two categorical variables are related**.

Example Dataset

Gender vs Product Preference

Gender Product A Product B

Male	40	30
Female	20	50

Question:

Is **product preference dependent on gender**?

Hypotheses

H_0 : Variables are independent

H_1 : Variables are related

Contingency Table

A **contingency table** summarizes frequencies of categorical variables.

Example:

	Product A	Product B	Total
Male	40	30	70
Female	20	50	70
Total	60	80	140

Expected value formula:

$$E = (\text{Row Total} \times \text{Column Total}) / \text{Grand Total}$$

Business Example

A retailer wants to check whether **customer gender influences product choice**.

If the chi-square test shows a relationship, marketing strategies can be customized.

Example use cases:

- Gender vs product preference
- Region vs subscription plan
- Age group vs product category
- Customer segment vs churn status

Five Worked Examples

Example 1

Gender vs product preference.

Example 2

Customer segment vs churn.

Example 3

Region vs purchase frequency.

Example 4

Education level vs employment status.

Example 5

Device type vs website usage.

Mini Project — Session 14

Project: Customer Preference Analysis

Dataset columns:

CustomerID

Gender

ProductCategory

Tasks

1. Create contingency table
2. Perform chi-square test
3. Calculate expected frequencies
4. Compute χ^2 statistic
5. Compute p-value
6. Interpret results

7. Write 5 business insights

10 Practice Questions

Theory (5)

1. What is the Chi-square test? : It is a statistical method used to check if there is a significant difference between expected and actual data.
2. When is Chi-square used? : It is used when dealing with categorical data to test relationships or differences between groups.
3. What is a contingency table? : It is a table that shows the frequency distribution of variables to analyze their relationship.
4. Define goodness of fit test. : It checks how well the observed data matches the expected data.
5. Define independence test. : It tests whether two categorical variables are related or independent of each other.

Practical (5)

1. Write χ^2 formula. : $\chi^2 = \sum [(O-E)^2 / E]$
2. How are expected frequencies calculated? : $EF = (\text{Row total} \times \text{Column total}) / \text{Grand total}$
3. If p-value < 0.05, what is the conclusion? : We reject the null hypothesis.
4. Give business example using Chi-square test. : A company uses it to check if customer preference depends on gender.
5. Identify variables suitable for Chi-square analysis. : Categorical variables like gender, color, type, or category are suitable.

SESSION 15 — ANOVA (Analysis of Variance)

ANOVA is widely used in analytics for comparing **multiple groups simultaneously**.

It is commonly used by companies like **Amazon**, **Netflix**, and **Google** when analyzing experiments involving **multiple variations**.

Example:

Testing **3 advertisement campaigns** instead of just A/B testing.

What Is ANOVA?

ANOVA (Analysis of Variance) is a statistical test used to determine whether the **means of three or more groups are significantly different**.

Instead of performing many t-tests (which increases error risk), ANOVA compares all groups **in one test**.

Example Scenario

A company tests **three marketing campaigns**.

Campaign Avg Sales

Campaign A 200

Campaign B 240

Campaign C 300

Question:

Are these differences due to **random variation** or **real marketing impact**?

ANOVA answers this.

Hypotheses in ANOVA

Null hypothesis:

$H_0: \mu_1 = \mu_2 = \mu_3$

Meaning all group means are equal.

Alternative hypothesis:

H_1 : At least one mean is different

Core Idea of ANOVA

ANOVA compares **two types of variation**:

Type	Meaning
Between-group variance	Differences between group means
Within-group variance	Variability inside each group

Logic

If between-group variance is **much larger** than within-group variance, the groups are likely **different**.

ANOVA Test Statistic (F-Test)

The ANOVA test uses the **F-statistic**.

$F = \text{Between Group Variance} / \text{Within Group Variance}$

Interpretation

F value	Meaning
Small	Groups are similar
Large	Groups differ significantly

Example — Customer Purchase Value

Three age groups:

Age Group Purchase Values

18–25 120, 140, 160

26–35 200, 210, 220

36–50 300, 320, 330

ANOVA determines whether these **age groups spend differently**.

Types of ANOVA

One-Way ANOVA

Used when **one factor** affects the outcome.

Example:

Marketing campaign → Sales

Two-Way ANOVA

Used when **two factors** affect outcome.

Example:

Marketing campaign + Region → Sales

Business Use Cases

ANOVA is used in many real-world analytics tasks.

Marketing Analysis

Compare performance of multiple ad campaigns.

Product Pricing

Compare revenue from different pricing strategies.

Customer Segmentation

Compare spending across customer groups.

Education Analytics

Compare exam scores across schools.

Manufacturing

Compare product quality across factories.

Five Worked Examples

Example 1

Compare sales of three marketing campaigns.

Example 2

Compare customer spending across age groups.

Example 3

Compare delivery time across three warehouses.

Example 4

Compare employee productivity across departments.

Example 5

Compare product ratings across brands.

Mini Project — Session 15

Project: Customer Spending by Age Group

Dataset columns:

CustomerID

AgeGroup

PurchaseValue

Tasks

1. Calculate mean purchase value for each age group
 2. Perform one-way ANOVA
 3. Calculate F-statistic
 4. Compute p-value
 5. Determine whether groups differ
 6. Visualize results
 7. Write 5 business insights
-

10 Practice Questions

Theory (5)

1. What is ANOVA? : It is a statistical test used to compare the means of three or more groups.
2. Why not use multiple t-tests instead? : Because multiple t-tests increase the chance of errors (false positives).

3. What is the F-statistic? : It is a value used in ANOVA to compare variance between groups and within groups.
4. What is between-group variance? : It shows how much the group means differ from each other.
5. What is within-group variance? : It shows how much the data varies inside each group.

Practical (5)

1. State ANOVA hypotheses. : Null hypothesis: all group means are equal; Alternative hypothesis: at least one mean is different.
2. When do we reject the null hypothesis? : When $p\text{-value} < 0.05$
3. Give business example using ANOVA. : A company uses ANOVA to compare sales performance across different regions.
4. What does large F-value indicate? : It indicates a significant difference between group means.
5. Difference between one-way and two-way ANOVA. : One-way ANOVA uses one factor, while two-way ANOVA uses two factors to analyze data.

SESSION 16 — Correlation Analysis

Correlation analysis helps us understand **how two variables move together**.

Data analysts use correlation to answer questions like:

- Does advertising increase sales?
- Does higher income lead to higher spending?
- Do longer sessions increase conversions?

Many companies such as **Amazon**, **Netflix**, and **Google** use correlation analysis to identify relationships between user behavior and business outcomes.

What Is Correlation?

Correlation measures the **strength and direction of relationship between two variables**.

It tells us whether:

- Variables move together
 - Variables move in opposite directions
 - Variables have no relationship
-

Correlation Coefficient

The correlation coefficient is usually represented by:

r

Range:

$-1 \leq r \leq 1$

Interpretation Table

r value Meaning

1	Perfect positive correlation
0.7	Strong positive relationship
0.3	Weak positive relationship
0	No correlation
-0.3	Weak negative relationship
-0.7	Strong negative relationship

r value Meaning

-1 Perfect negative correlation

Example

Advertising vs Sales

If:

$r = 0.85$

Then advertising strongly increases sales.

Types of Correlation

There are two main types used in analytics.

Pearson Correlation

Spearman Correlation

Pearson Correlation

Pearson correlation measures **linear relationships between numerical variables**.

Formula

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

Where:

- x, y = variables
 - $\bar{x}, \bar{y}$ = means
-

Example

Relationship between:

Advertising spend → Sales revenue

Conditions

Pearson works best when:

- Relationship is linear
- Data is normally distributed

- Variables are continuous
-

Spearman Correlation

Spearman correlation measures **monotonic relationships using ranks**.

It works even when the relationship is **non-linear**.

Example

Customer satisfaction rank vs loyalty rank.

When to Use

Spearman is useful when:

- Data is ordinal
 - Data has outliers
 - Relationship is monotonic but not linear
-

Correlation Matrix

When analyzing many variables, we use a **correlation matrix**.

Example:

Variable Sales Ads Price

Sales 1 0.8 -0.5

Ads 0.8 1 -0.2

Price -0.5 -0.2 1

Interpretation

Sales and Ads → strong positive relationship

Sales and Price → negative relationship

Correlation Heatmap

A heatmap visually shows correlations between variables.

Colors represent correlation strength:

- Dark color → strong correlation

- Light color → weak correlation

Five Business Examples

Example 1

Advertising spend vs sales revenue.

Example 2

Customer income vs purchase amount.

Example 3

Website session time vs conversion rate.

Example 4

Product price vs demand.

Example 5

Customer satisfaction vs retention.

Common Pitfalls

Correlation ≠ Causation

Just because two variables correlate **does not mean one causes the other**.

Example:

Ice cream sales and drowning incidents both increase in summer.

They correlate but do not cause each other.

Mini Project — Session 16

Project: Sales Drivers Analysis

Dataset columns:

Sales

AdvertisingSpend

ProductPrice

CustomerVisits

Tasks

1. Load dataset
 2. Compute correlation matrix
 3. Visualize heatmap
 4. Identify strongest positive relationship
 5. Identify strongest negative relationship
 6. Write 5 business insights
-

10 Practice Questions

Theory (5)

1. What is correlation? : Correlation is a measure of how two variables are related to each other.
 2. What is correlation coefficient? : It is a number (between -1 and +1) that shows the strength and direction of the relationship.
 3. Difference between Pearson and Spearman correlation. : Pearson measures linear relationships, Spearman measures rank-based relationships.
 4. What does $r = -0.8$ mean? : It means a strong negative relationship, where one variable increases as the other decreases.
 5. Explain correlation vs causation. : Correlation shows a relationship, but causation means one variable actually causes the other.
-

Practical (5)

1. Compute correlation between two variables. : Correlation is calculated using a formula (like Pearson) that compares how two variables change together.
2. Interpret correlation matrix. : A correlation matrix shows relationships between multiple variables, where values close to +1 or -1 indicate strong relationships.
3. Explain heatmap visualization. : A heatmap is a color-coded chart used to visually show the strength of relationships in a correlation matrix.
4. Give example of negative correlation. : As price increases, demand decreases is an example of negative correlation.
5. Give real business example of correlation analysis. : A company analyzes the relationship between advertising spend and sales to see if higher ads lead to higher sales.

SESSION 17 — Introduction to Regression

(Simple Linear Regression)

Regression is one of the **most widely used predictive techniques in analytics**.

Companies like **Amazon**, **Google**, and **Netflix** use regression models to predict:

- Sales
 - Demand
 - Customer lifetime value
 - Advertising impact
 - Pricing effects
-

What Is Regression?

Regression analysis is a statistical method used to **model the relationship between a dependent variable and one or more independent variables**.

In simple words:

Regression helps us **predict one variable using another variable**.

Example

Predict:

Sales using Advertising Spend.

Advertising → Independent variable

Sales → Dependent variable

Simple Linear Regression

Simple linear regression models the relationship between **two variables using a straight line**.

Regression Equation

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Where:

- Y = Dependent variable
- X = Independent variable
- β_0 = Intercept
- β_1 = Slope

- ϵ = Error term

Line of Best Fit

Regression finds the **line that best fits the data points**.

The best line minimizes the **sum of squared errors**.

The line shows how the dependent variable changes when the independent variable changes.

Interpreting Regression Coefficients

Intercept (β_0)

Value of Y when X = 0.

Example:

Base sales when advertising spend is zero.

Slope (β_1)

Represents **change in Y for a unit change in X**.

Example:

If:

$$\beta_1 = 5$$

Then every ₹1 increase in advertising increases sales by ₹5.

Example Dataset

Advertising vs Sales

Advertising Sales

10	100
20	150
30	200
40	250

Regression model might be:

$$\text{Sales} = 50 + 5 \times \text{Advertising}$$

Meaning:

Every ₹1 increase in advertising increases sales by ₹5.

R-Squared (Model Accuracy)

R² (Coefficient of Determination) measures how well the regression model explains the data.

Range

$$0 \leq R^2 \leq 1$$

Interpretation

R² Meaning

0 Model explains nothing

0.5 Explains 50% of variation

0.8 Strong model

1 Perfect prediction

Real Business Examples

Example 1 — Marketing

Advertising spend → Sales revenue.

Example 2 — Pricing

Product price → Demand.

Example 3 — Real Estate

House size → House price.

Example 4 — Website Analytics

Session duration → Conversion probability.

Example 5 — Finance

Investment amount → Returns.

Assumptions of Linear Regression

For regression to work well:

1. Linear relationship between variables
 2. Independent observations
 3. Homoscedasticity (constant variance)
 4. Normally distributed errors
-

Five Worked Examples

Example 1

Predict sales using advertising.

Example 2

Predict house price using house size.

Example 3

Predict revenue using customer traffic.

Example 4

Predict electricity usage using temperature.

Example 5

Predict delivery time using distance.

Mini Project — Session 17

Project: Advertising Impact on Sales

Dataset columns:

AdvertisingSpend

Sales

Tasks

1. Load dataset
2. Plot scatter plot
3. Train linear regression model

4. Interpret coefficients
 5. Compute R^2 score
 6. Predict future sales
 7. Write 5 business insights
-

10 Practice Questions

Theory (5)

1. What is regression? : Regression is a statistical method used to model and analyze the relationship between a dependent variable and one or more independent variables.
 2. Define dependent variable. : The dependent variable is the outcome or result that we want to predict or explain.
 3. Define independent variable. : The independent variable is the factor that influences or is used to predict the dependent variable.
 4. What does slope represent? : The slope represents the rate of change, showing how much the dependent variable changes for a one-unit increase in the independent variable.
 5. What is R^2 ? : R^2 (coefficient of determination) shows how much of the variation in the dependent variable is explained by the model.
-

Practical (5)

1. Write regression equation. : $Y = \beta_0 + \beta_1 X + \epsilon$
2. Interpret β_1 coefficient. : It shows how much the dependent variable changes for a one-unit increase in the independent variable.
3. What does $R^2 = 0.8$ mean? : It means 80% of the variation in the dependent variable is explained by the model.
4. Give business example using regression. : A company uses regression to predict sales based on advertising spend.
5. Identify dependent and independent variables in a dataset. : Dependent variable is the output (target), while independent variables are the inputs used to predict it.

SESSION 1 — Introduction to Linear Algebra (Vector Basics)

What is Linear Algebra?

Linear Algebra is the branch of mathematics that studies **vectors, matrices, and linear transformations**.

It provides the mathematical language to describe:

- Data structures
 - Machine learning models
 - Optimization algorithms
-

Why Linear Algebra Matters in Machine Learning

Almost every ML algorithm can be written using **matrix and vector operations**.

Examples:

ML Task	Linear Algebra Use
Linear Regression	Matrix equations
Neural Networks	Weight matrices
PCA	Eigenvectors
Recommendation Systems	Matrix factorization
Computer Vision	Image matrices

Understanding Vectors

A **vector** is an ordered list of numbers that represents a point in space.

Example:

$v=[2,4,6]$

This vector has **3 dimensions**.

Dimension of a Vector

The number of elements in a vector is called its **dimension**.

Example:

Vector Dimension

[5, 8] 2D

[1, 3, 5] 3D

[2, 4, 6, 8] 4D

Representing Data as Vectors

In data science, **each observation can be represented as a vector**.

Example dataset:

Age Income Spending

25 40 200

30 50 250

Customer vector:

$x=[25,40,200]$

This is a **feature vector**.

Vector Addition

Formula

$a+b=[a_1+b_1, a_2+b_2]$

Example

$a=[2,4]$ $b=[1,3]$, $a+b=[3,7]$

Vector Subtraction

$a-b=[a_1-b_1, a_2-b_2]$

Example:

$[5,7]-[2,3]=[3,4]$

Scalar Multiplication

Multiplying a vector by a number.

$k \times v=[kv_1, kv_2]$

Example:

$$3 \times [2, 4] = [6, 12]$$

Dot Product

Definition

The **dot product** multiplies corresponding elements and sums them.

Formula

$$a \cdot b = a_1b_1 + a_2b_2 + \dots + a_nb_n$$

Example

$$a = [2, 3] \quad b = [4, 5]$$

$$a \cdot b = (2 \times 4) + (3 \times 5)$$

$$8 + 15 = 23$$

Geometric Interpretation

Dot product also relates to **angle between vectors**.

$$a \cdot b = |a||b|\cos(\theta)$$

Meaning:

- If vectors point in same direction $\rightarrow$ large dot product
 - If perpendicular $\rightarrow$ dot product = 0
-

Vector Norms

Vector norm measures **vector length (magnitude)**.

L1 Norm (Manhattan Distance)

Formula:

$$||x||_1 = |x_1| + |x_2| + \dots + |x_n|$$

Example:

$$= L1 = |3| + |4| = 7$$

L2 Norm (Euclidean Distance)

Formula:

$$||x||_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

Example:

$$x=[3,4] \quad ||x||_2 = \sqrt{3^2 + 4^2} = 5$$

Distance Between Vectors

Euclidean distance:

$$d = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

Example:

Points:

$$A = [1, 2]$$

$$B = [4, 6]$$

Distance:

$$\sqrt{(4-1)^2 + (6-2)^2}$$

$$\sqrt{9 + 16} = 5$$

Cosine Similarity (Intro)

Measures similarity between vectors.

Formula:

$$\cos(\theta) = \frac{A \cdot B}{||A|| ||B||}$$

Range:

Value Meaning

1 identical direction

0 unrelated

-1 opposite direction

Used heavily in:

- Text similarity
- Recommendation systems
- Search engines

Example:

Movie recommendation systems like **Netflix** compare user preference vectors using cosine similarity.

Mini Project — Session 1

Dataset:

Customer features:

Age, Income, SpendingScore

Tasks:

1. Represent each customer as vector
2. Compute vector addition
3. Compute dot product between customers
4. Compute L2 norm
5. Calculate distance between two customers
6. Compute cosine similarity

Goal:

Identify **similar customers for recommendation systems**.

Practice Questions

Theory (5)

1. What is linear algebra? : Linear algebra is a branch of mathematics that deals with vectors, matrices, and linear equations.
 2. Define vector. : A vector is an ordered list of numbers representing magnitude and direction.
 3. What is vector dimension? : Vector dimension is the number of elements (components) in the vector.
 4. What is dot product? : Dot product is the sum of the products of corresponding elements of two vectors.
 5. Difference between L1 and L2 norm. : L1 norm is the sum of absolute values, while L2 norm is the square root of the sum of squares.
-

Practical (5)

1. Add vectors [2,3] and [4,5]. : [6,8]
2. Compute dot product of [1,2] and [3,4]. : $3 + 8 = 11$
3. Compute L2 norm of [3,4]. : 5

4. Compute distance between [1,2] and [4,6]. : 5
5. Compute cosine similarity conceptually. : It is the dot product divided by the product of magnitudes, measuring the angle between two vectors.

SESSION 2 — Matrices (Part 1: Foundations)

Matrices are fundamental for representing and manipulating datasets in machine learning. Large platforms like **Google** and **Netflix** process billions of rows of data internally using matrix operations.

What Is a Matrix?

A **matrix** is a rectangular array of numbers arranged in **rows and columns**.

Example:

A= [123
456]

This matrix has:

- 2 rows
 - 3 columns
-

Rows and Columns

Term	Meaning
------	---------

Row	Horizontal elements
-----	---------------------

Column	Vertical elements
--------	-------------------

Example matrix:

[10, 20]
[30, 40]

Rows:

[10, 20]
[30, 40]

Columns:

[10, 30]
[20, 40]

Matrix Shape

The **shape** of a matrix is defined as:

rows×column

Example:

Matrix **Shape**

2 rows, 3 columns 2×3

3 rows, 2 columns 3×2

Example matrix:

A= [5 8 2

3 7 9]

Shape = 2×3

Matrix Notation

Matrices are usually represented using capital letters.

Example:

A= [12

34]

Elements are written as:

A_{ij}

Where:

- i = row index
- j = column index

Example:

$A_{12}=2$

Representing Datasets as Matrices

In data science, **datasets are stored as matrices.**

Example dataset:

Age Income SpendingScore

25 40 200

30 50 250

35 60 300

Matrix form:

X= [25 40 200

30 50 250

35 60 300]

Rows = observations

Columns = features

Matrix Addition

Matrices can be added **only if they have the same shape.**

Formula

$A+B = [a_{11}+b_{11} \ a_{12}+b_{12}]$

Example

A= [1 2

3 4]

B= [5 6

7 8]

A+B= [6 8

10 12]

Matrix Subtraction

Same rule: matrices must have **same dimensions.**

Scalar Multiplication of Matrices

Multiplying each element of a matrix by a constant.

Example:

$3 \times [1 \ 2$

3 4]

Result:

[3 6

9 12]

Real-World Example — Dataset Matrix

Customer dataset:

Age Income Spending

25 40 200

30 50 250

Matrix:

X= [25 40 200

30 50 250]

Machine learning models treat **this entire dataset as a matrix**.

Mini Project — Session 2

Dataset:

Customer information:

Age Income Spending

22 30 150

25 35 200

30 50 250

Tasks:

1. Represent dataset as matrix
2. Identify matrix shape
3. Multiply matrix by scalar (2×)
4. Add two matrices
5. Interpret rows and columns

Goal:

Understand how datasets become matrices for ML.

Theory (5)

1. What is a matrix? : A matrix is a rectangular arrangement of numbers organized in rows and columns.
 2. What is matrix shape? : Matrix shape tells the number of rows and columns, written as (rows $\times$ columns).
 3. What does A_{ij} represent? : It represents the element present in the i -th row and j -th column of a matrix.
 4. When can matrices be added? : Matrices can be added only when they have the same shape (same rows and columns).
 5. How are datasets stored as matrices? : Datasets are stored as matrices where rows represent observations and columns represent features.
-

Practical (5)

1. Identify shape of matrix : 2×3

[123

456]

Add two matrices : [5 7 9]

2. Subtract matrices : [-3 -3 -3]
3. Multiply matrix by scalar : [2 4 6], [8 10 12]
4. Convert dataset into matrix form : if (1,2),(3,4) then [1 2], [3 4]

SESSION 3 — Matrices (Part 2: Multiplication & Operations)

Matrix multiplication is the **core mathematical operation behind machine learning models**, neural networks, and recommendation systems.

Large-scale ML systems at companies like **Google**, **Amazon**, and **Netflix** rely heavily on matrix multiplications to process billions of data points.

Matrix Multiplication

Matrix multiplication combines two matrices to produce a new matrix.

However, multiplication is only possible when the **dimensions are compatible**.

Dimension Compatibility Rule

If:

Matrix A $\rightarrow m \times n$

Matrix B $\rightarrow n \times p$

Then:

$A \times B = C$

Where:

$C = m \times p$

Example

Matrix A:

2×3

Matrix B:

3×4

Result:

2×4

Row $\times$ Column Concept

Matrix multiplication works by:

Row of first matrix $\times$ Column of second matrix.

Example

A = $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ B = $\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$

Step 1

First row of A $\times$ first column of B

$(1 \times 5) + (2 \times 7) = 5 + 14 = 19$

Step 2

First row $\times$ second column

$(1 \times 6) + (2 \times 8) = 6 + 16 = 22$

Step 3

Second row $\times$ first column

$$(3 \times 5) + (4 \times 7) = 15 + 28 = 43$$

Step 4

Second row $\times$ second column

$$(3 \times 6) + (4 \times 8) = 18 + 32 = 50$$

Final Result

$$A \times B = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Identity Matrix

The **identity matrix** is a square matrix where:

- Diagonal elements = 1
 - All other elements = 0
-

Example:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Property

Multiplying any matrix by identity matrix keeps it unchanged.

$$A \times I = A$$

Zero Matrix

A **zero matrix** contains only zeros.

Example:

$$Z = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

0 0]

Property:

$$A \times Z = Z$$

Matrix Transpose

Transpose of a matrix flips **rows into columns**.

Notation: A^T

Example

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 5 & 6 \end{bmatrix}$$

Transpose:

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 6 \end{bmatrix}$$

Real-World Example — Dataset Transformation

Suppose we have a dataset:

Feature1 Feature2

2 3

4 5

Matrix:

$$X = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 5 \end{bmatrix}$$

Suppose we apply a transformation matrix:

$$W = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2 \end{bmatrix}$$

Then:

$$XW$$

This transforms features — something ML models do internally.

Where Matrix Multiplication Is Used

Matrix multiplication is used in:

Application	Example
Neural Networks	Weight $\times$ input
Linear Regression	$X^T X$ operations
Computer Vision	Image transformations
Recommendation Systems	User $\times$ product matrix
Natural Language Processing	Word embeddings

Mini Project — Session 3

Dataset:

Customer features:

Age Income

25 40

30 50

35 60

Tasks:

1. Represent dataset as matrix
2. Create transformation matrix
3. Perform matrix multiplication
4. Compute transpose of dataset
5. Interpret results

Goal:

Understand how ML models transform features.

Theory (5)

1. What is matrix multiplication? : Matrix multiplication is the process of multiplying rows of one matrix with columns of another to produce a new matrix.
2. What is dimension compatibility rule? : Matrix multiplication is possible only when the number of columns of the first matrix equals the number of rows of the second matrix.

3. What is identity matrix? : An identity matrix is a square matrix with 1s on the diagonal and 0s at all other places.
 4. What is zero matrix? : A zero matrix is a matrix where all elements are 0.
 5. What is matrix transpose? : Matrix transpose is obtained by converting rows into columns and columns into rows.
-

Practical (5)

1. Multiply two matrices
2. Check compatibility for multiplication
3. Find transpose of matrix
4. Multiply matrix by identity matrix
5. Identify shape of result matrix

SESSION 4 — Matrix Inverse & Determinant

This session explains how we solve **systems of linear equations using matrices**.

This is exactly how algorithms such as regression are solved mathematically.

Large ML systems at companies like **Google, Amazon, and Netflix** internally use similar mathematical foundations.

Determinant of a Matrix

The **determinant** is a numerical value calculated from a square matrix.

It tells us important properties of the matrix such as:

- Whether the matrix has an inverse
 - Whether equations can be solved uniquely
-

Determinant Notation

For matrix A:

$\det(A)$ or $|A|$

Determinant of a 2x2 Matrix

Matrix:

A = $\begin{bmatrix} a & b \end{bmatrix}$

C d]

Determinant formula:

$$|A|=ad-bc$$

Example

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

$$|A| = (2 \times 5) - (3 \times 4)$$

$$= 10 - 12 = -2$$

Determinant Meaning

The determinant indicates **whether the matrix is invertible**.

Determinant Meaning

0 Matrix has no inverse

Non-zero Matrix has inverse

Matrix Inverse

The **inverse of a matrix** is another matrix that reverses the effect of multiplication.

Notation:

$$A^{-1}$$

Property

$$A \times A^{-1} = I$$

Where I is the identity matrix.

Inverse of a 2x2 Matrix

Matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Inverse formula:

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Example

$$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$$

Step 1: determinant

$$(2 \times 3) - (1 \times 5) = 6 - 5 = 1$$

Step 2: inverse

$$A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$$

When Does an Inverse Exist?

A matrix has an inverse **only if**:

Matrix is square

Determinant $\neq 0$

Singular Matrix

A matrix with **determinant = 0** is called a **singular matrix**.

Properties:

- No inverse exists
- Linear system cannot be solved uniquely

Example:

$$A = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$$

Determinant:

$$(2 \times 2) - (4 \times 1) = 4 - 4 = 0$$

So this matrix is **singular**.

Solving Linear Equations Using Matrices

Consider the system:

$$Ax = b$$

Where:

A = coefficient matrix

x = unknown variables

b = output vector

Example:

$$2x + y = 5 \quad x + 3y = 6$$

Matrix form:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \quad x = \begin{bmatrix} x \\ y \end{bmatrix} \quad b = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

Solution Formula

$$x = A^{-1}b$$

This is how the system is solved.

Application in Linear Regression

The analytical solution of linear regression is:

$$\beta = (X^T X)^{-1} X^T y$$

Where:

- X = feature matrix
- y = target vector
- β = regression coefficients

This formula uses:

- Matrix transpose
- Matrix multiplication
- Matrix inverse

Mini Project — Session 4

Dataset:

Two linear equations:

$$3x + 2y = 16 \quad x + 4y = 14$$

Tasks:

1. Convert to matrix form $Ax = b$
2. Compute determinant
3. Find matrix inverse
4. Compute solution $x = A^{-1}b$

5. Interpret results

Goal:

Understand how ML models solve linear equations.

Theory (5)

1. What is determinant? : The determinant is a scalar value of a square matrix that indicates its properties like invertibility.
 2. What is matrix inverse? : The inverse of a matrix is another matrix that, when multiplied with the original, gives the identity matrix.
 3. When does inverse exist? : An inverse exists only if the determinant of the matrix is not zero.
 4. What is singular matrix? : A singular matrix is a matrix whose determinant is zero and has no inverse.
 5. Explain equation $Ax=b$: A = coefficient matrix, x = variable, and b = result.
-

Practical (5)

1. Compute determinant of 2×2 matrix
2. Identify singular matrix
3. Compute inverse of matrix
4. Solve system using matrix inverse
5. Explain regression equation using matrices